CHE 305 Chemistry

Gases

MODULE 5

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Characteristics of Gases

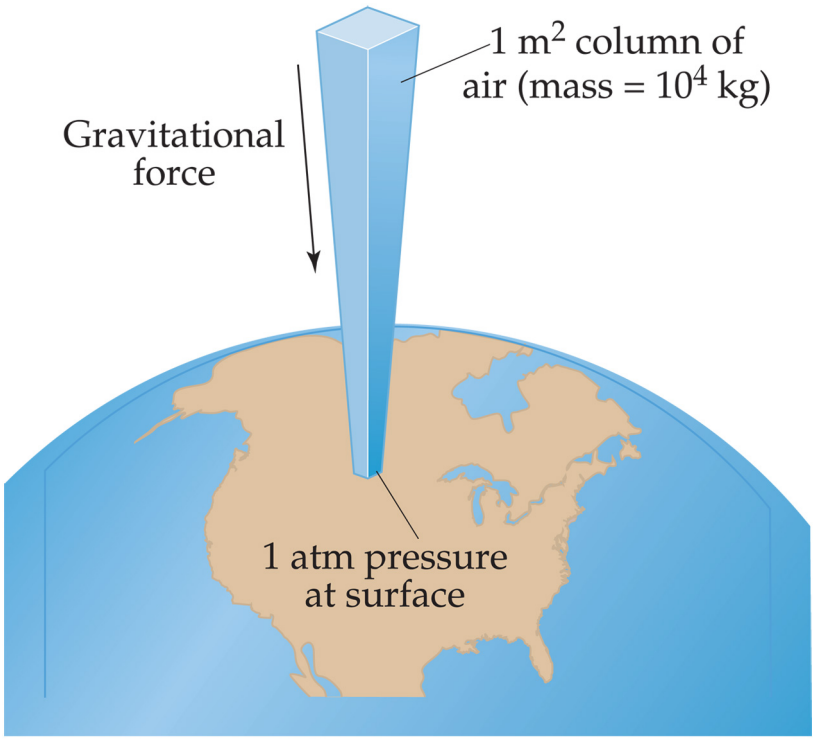
- Unlike liquids and solids, gases
  - expand to fill their containers;
  - are highly compressible;
  - have extremely low densities.

Pressure

- Pressure is the amount of force applied to an area.

$$P = \frac{F}{A}$$

- Atmospheric pressure is the weight of air per unit of area.

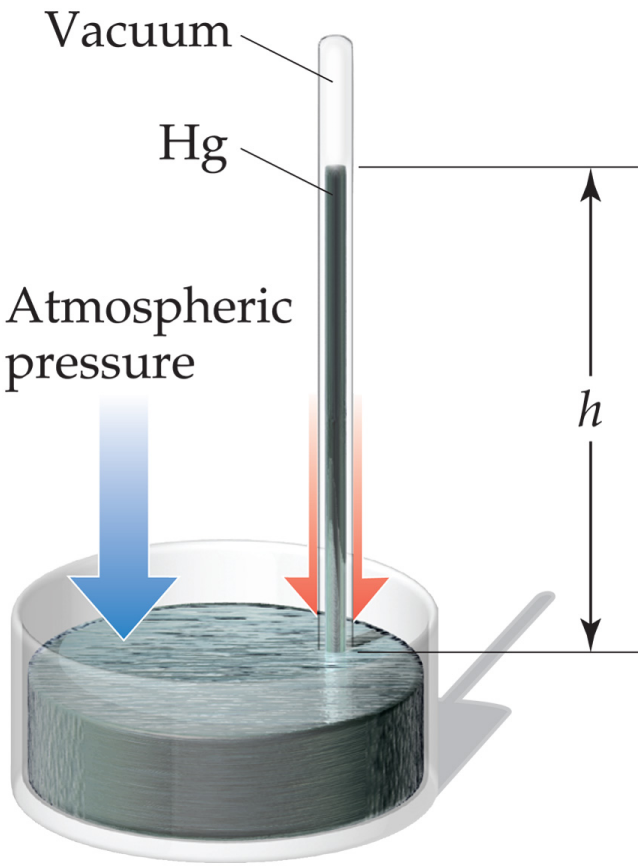


Units of Pressure

- Pascals
  - 1 Pa = 1 N/m^2
- Bar
  - 1 bar = 10^5 Pa = 100 kPa

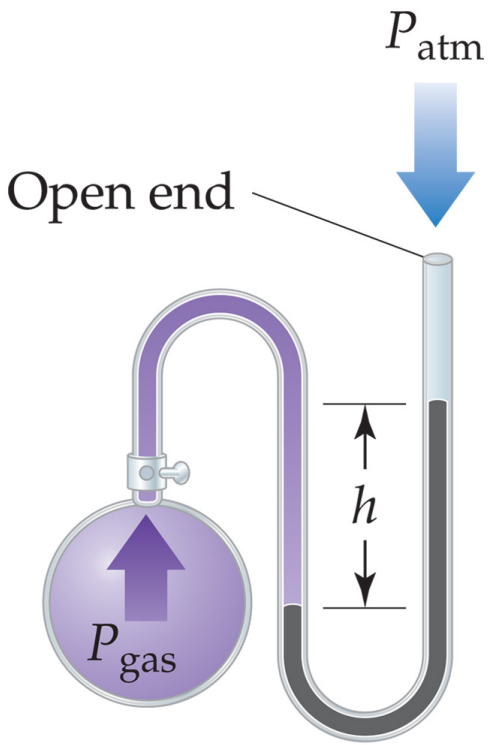
Units of Pressure

- mm Hg or torr
  - These units are literally the difference in the heights measured in mm (h) of two connected columns of mercury.



- Atmosphere
  1.00 atm = 760 torr

Manometer



$$P_{gas} = P_{atm} + P_h$$

This device is used to measure the difference in pressure between atmospheric pressure and that of a gas in a vessel.

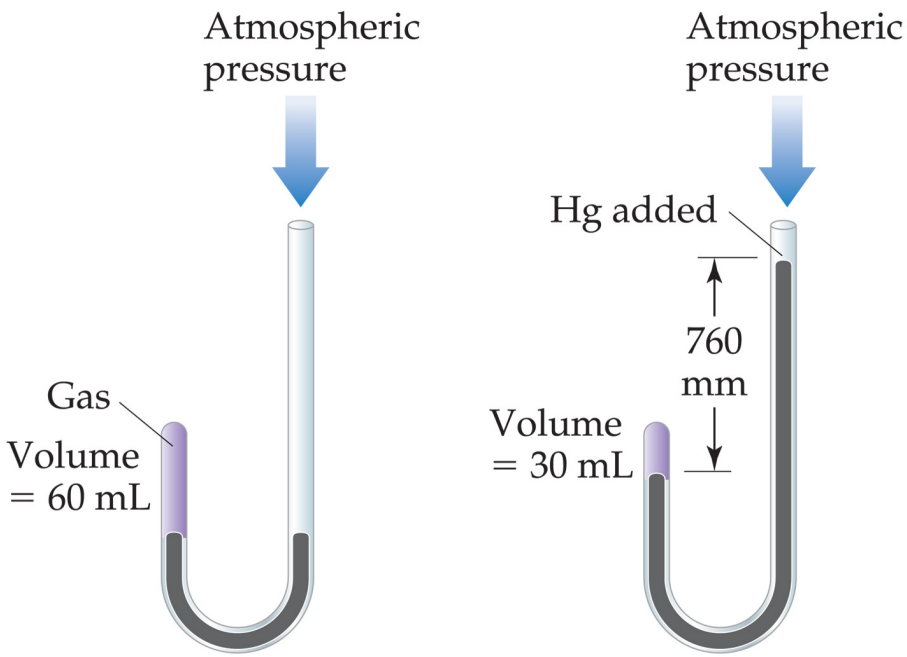
Standard Pressure

- Normal atmospheric pressure at sea level is referred to as standard pressure.

- It is equal to
  - 1.00 atm
  - 760 torr (760 mm Hg)
  - 101.325 kPa

Boyle's Law

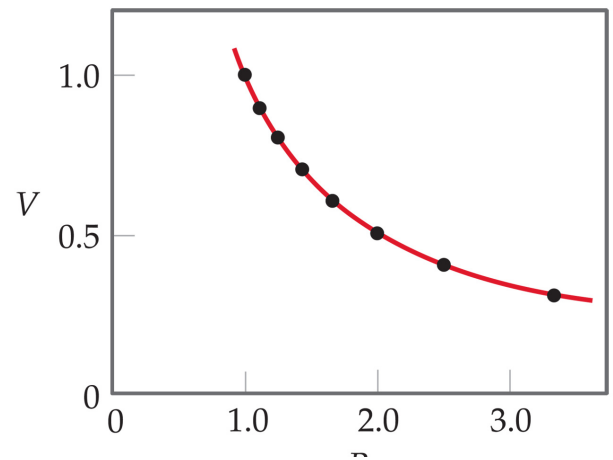
The volume of a fixed quantity of gas at constant temperature is inversely proportional to the pressure.



Gases

As *P* and *V* are inversely proportional

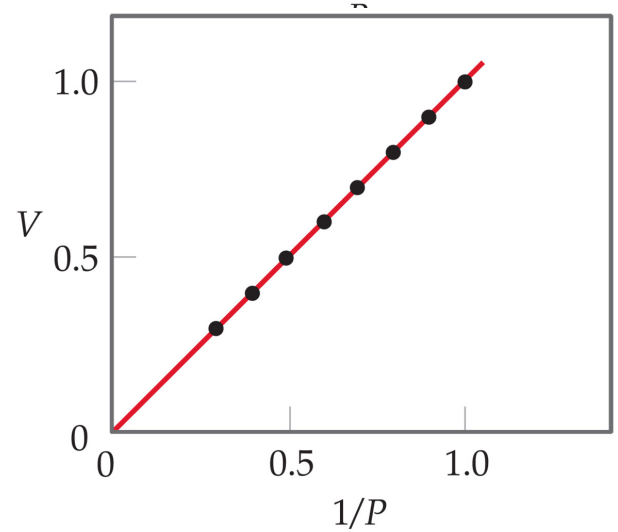
A plot of *V* versus *P* results in a curve.



Since $PV = k$

$V = k\,(1/P)$

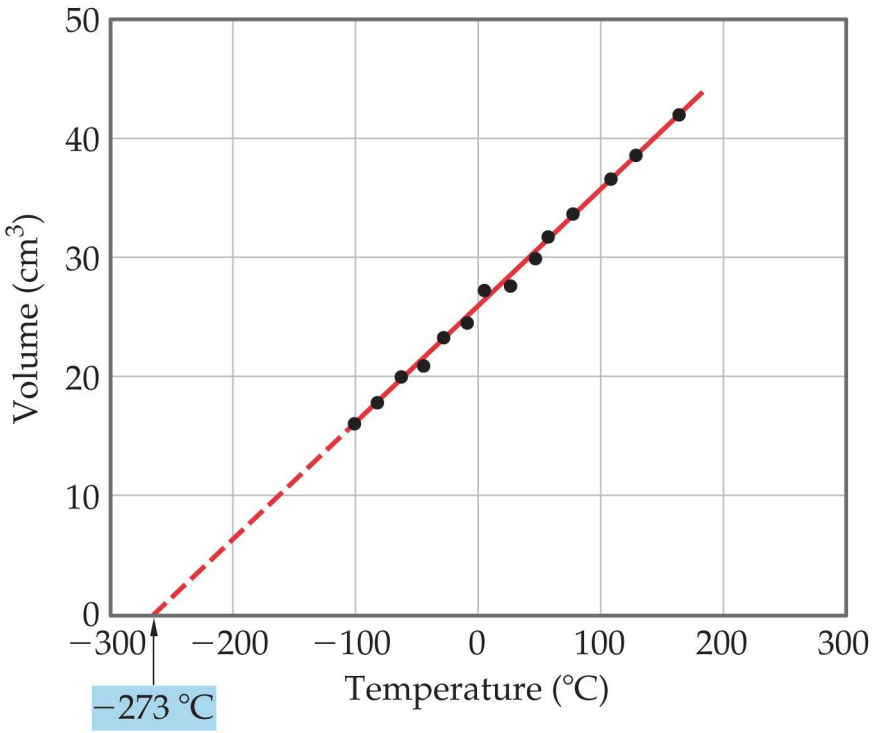
This means a plot of *V* versus 1/*P* will be a straight line.



Charles's Law

- The volume of a fixed amount of gas at constant pressure is directly proportional to its absolute temperature.

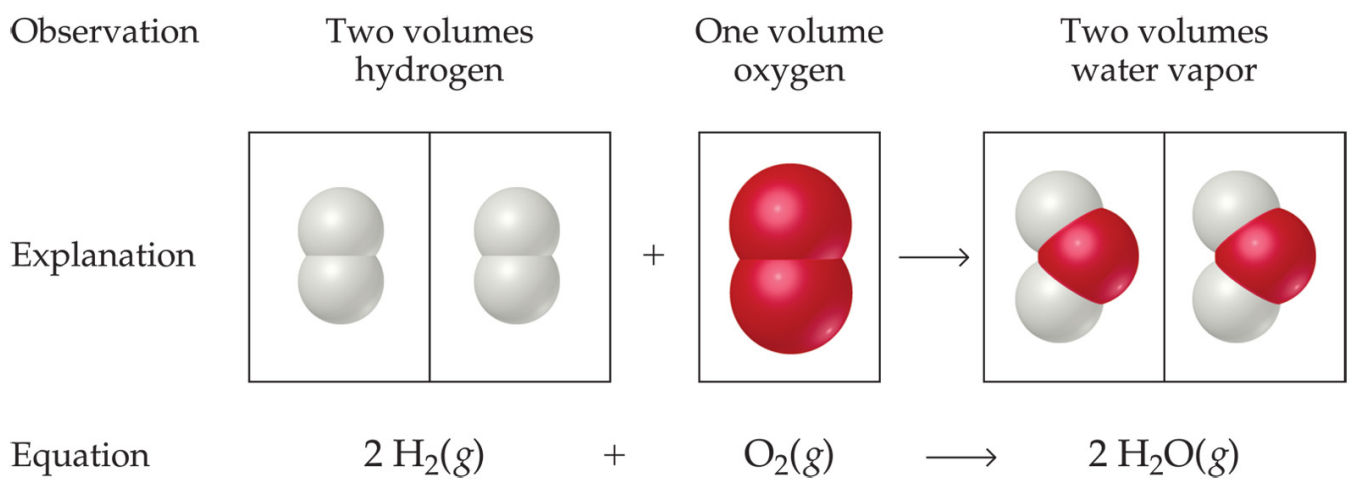
- i.e., $\frac{V}{T} = k$



A plot of *V* versus *T* will be a straight line.

Avogadro's Law

- The volume of a gas at constant temperature and pressure is directly proportional to the number of moles of the gas.
- Mathematically, this means $V = kn$



Gases

Ideal-Gas Equation

- So far we've seen that

$V \propto 1/P$ (Boyle's law)
$V \propto T$ (Charles's law)
$V \propto n$ (Avogadro's law)

- Combining these, we get

$$V \propto \frac{nT}{P}$$

Ideal-Gas Equation

The constant of proportionality is known as *R*, the gas constant.

Units	Numerical Value
L-atm/mol-K	0.08206
J/mol-K*	8.314
cal/mol-K	1.987
m^3-Pa/mol-K*	8.314
L-torr/mol-K	62.36

*SI unit

Ideal-Gas Equation

The relationship

$$V \propto \frac{nT}{P}$$

then becomes

$$V = R\,\frac{nT}{P}$$

or

$$PV = nRT$$

Densities of Gases

If we divide both sides of the ideal-gas equation by *V* and by *RT*, we get

$$\frac{n}{V} = \frac{P}{RT}$$

Densities of Gases

- We know that
  - moles × molecular mass = mass

$$n \times M = m$$

- So multiplying both sides by the molecular mass (M) gives

$$\frac{m}{V} = \frac{PM}{RT}$$

Densities of Gases

- Mass ÷ volume = density
- So, $$d = \frac{m}{V} = \frac{PM}{RT}$$

Note: One only needs to know the molecular mass, the pressure, and the temperature to calculate the density of a gas.

Molecular Mass

We can manipulate the density equation to enable us to find the molecular mass of a gas:

$$d = \frac{PM}{RT}$$

Becomes

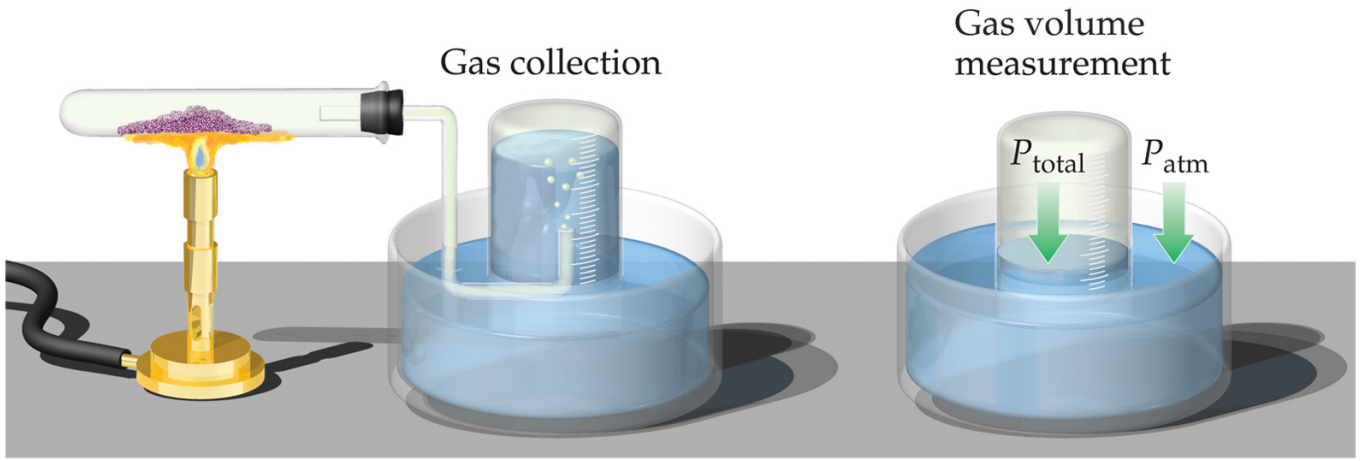
$$M = \frac{dRT}{P}$$

Dalton's Law of Partial Pressures

- The total pressure of a mixture of gases equals the sum of the pressures that each would exert if it were present alone.
- In other words,

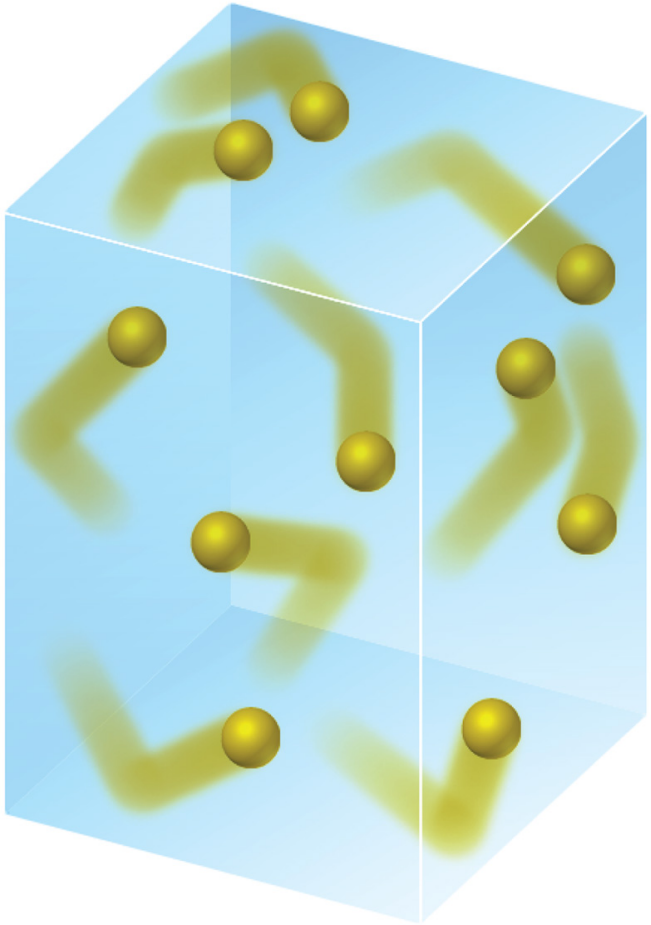
$$P_{total} = P_1 + P_2 + P_3 + \ldots$$

Partial Pressures



- When one collects a gas over water, there is water vapor mixed in with the gas.
- To find only the pressure of the desired gas, one must subtract the vapor pressure of water from the total pressure.

Kinetic-Molecular Theory



This is a model that aids in our understanding of what happens to gas particles as environmental conditions change.

Main Tenets of Kinetic-Molecular Theory

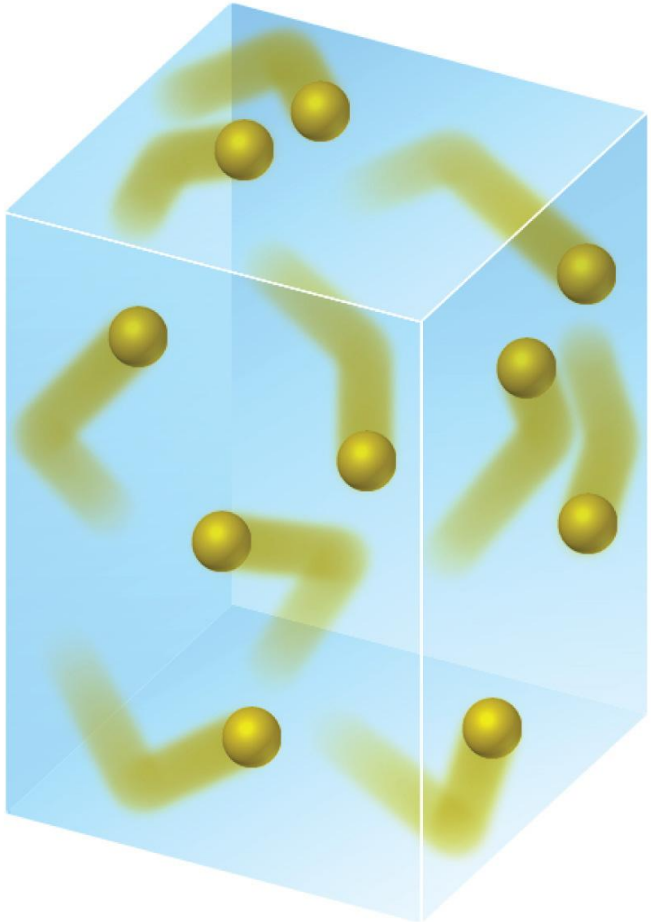
Gases consist of large numbers of molecules that are in continuous, random motion.

Main Tenets of Kinetic-Molecular Theory

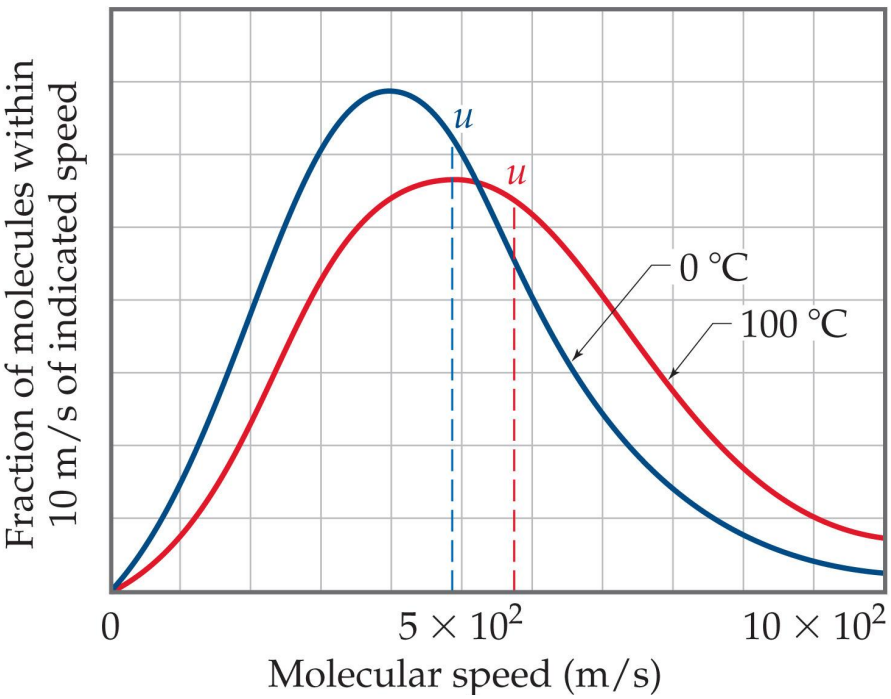
The combined volume of all the molecules of the gas is negligible relative to the total volume in which the gas is contained.

Main Tenets of Kinetic-Molecular Theory

Attractive and repulsive forces between gas molecules are negligible.



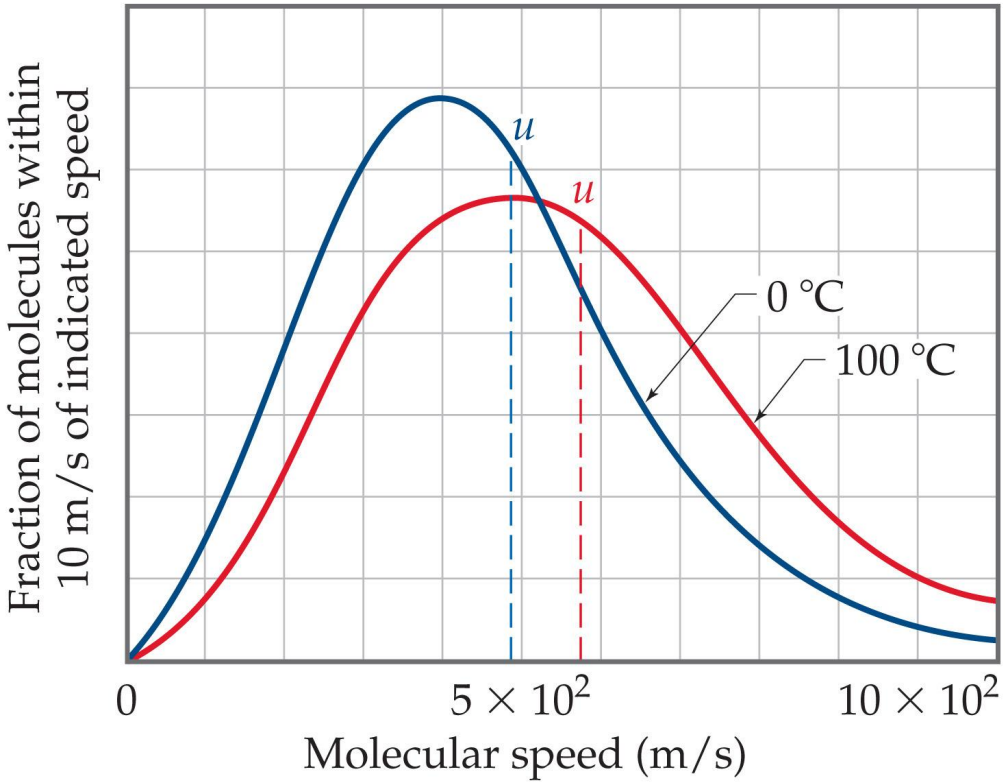
Main Tenets of Kinetic-Molecular Theory



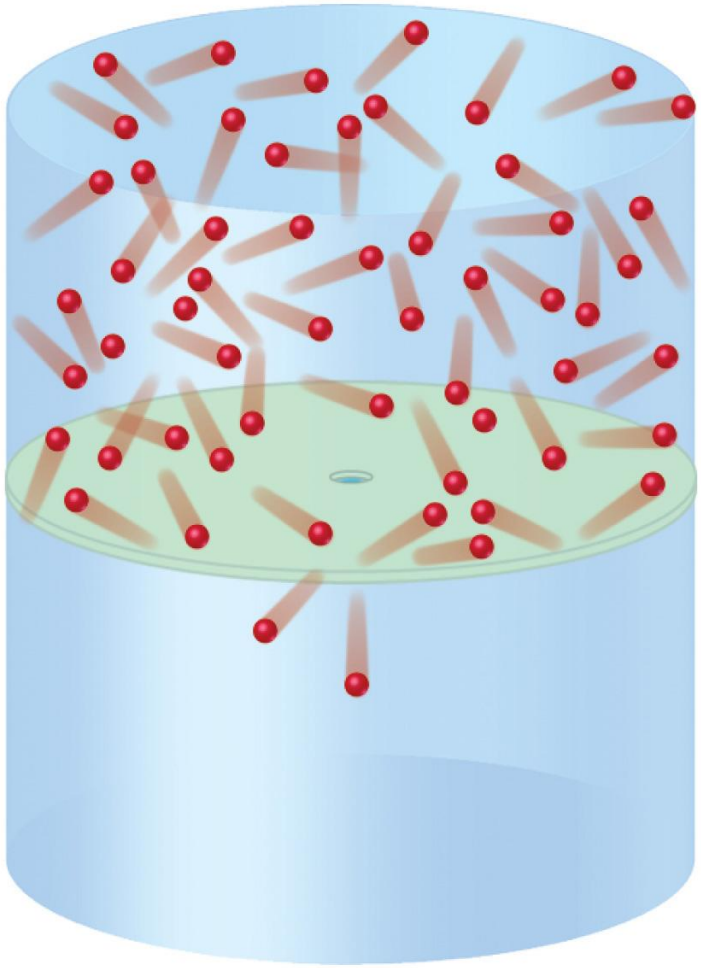
Energy can be transferred between molecules during collisions, but the *average* kinetic energy of the molecules does not change with time, as long as the temperature of the gas remains constant.

Main Tenets of Kinetic-Molecular Theory

The average kinetic energy of the molecules is proportional to the absolute temperature.



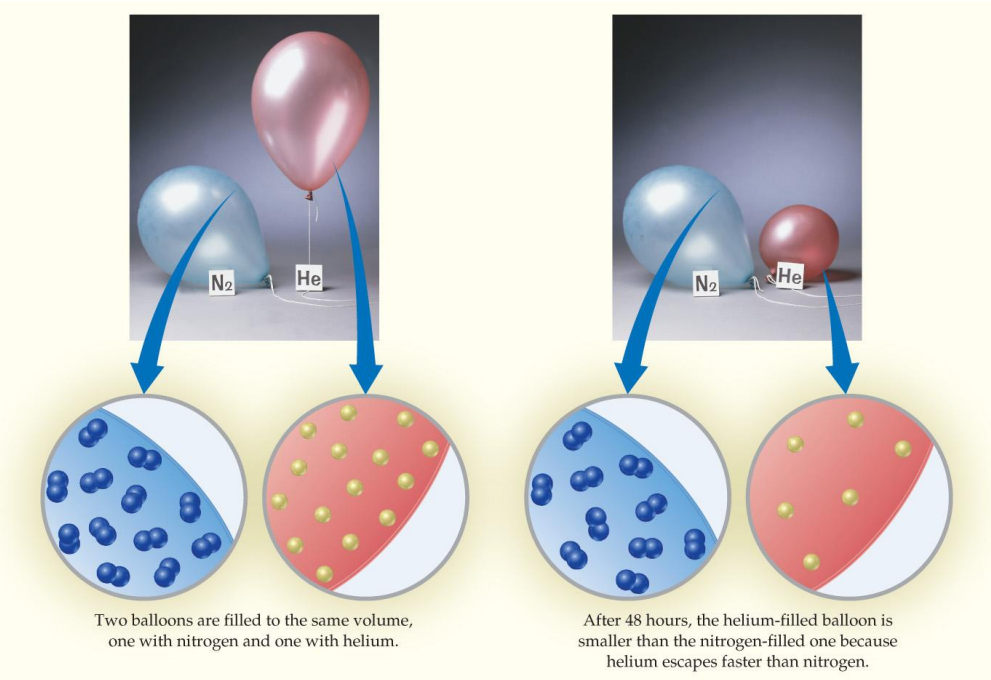
Effusion



Effusion is the escape of gas molecules through a tiny hole into an evacuated space.

Effusion

The difference in the rates of effusion for helium and nitrogen, for example, explains a helium balloon would deflate faster.

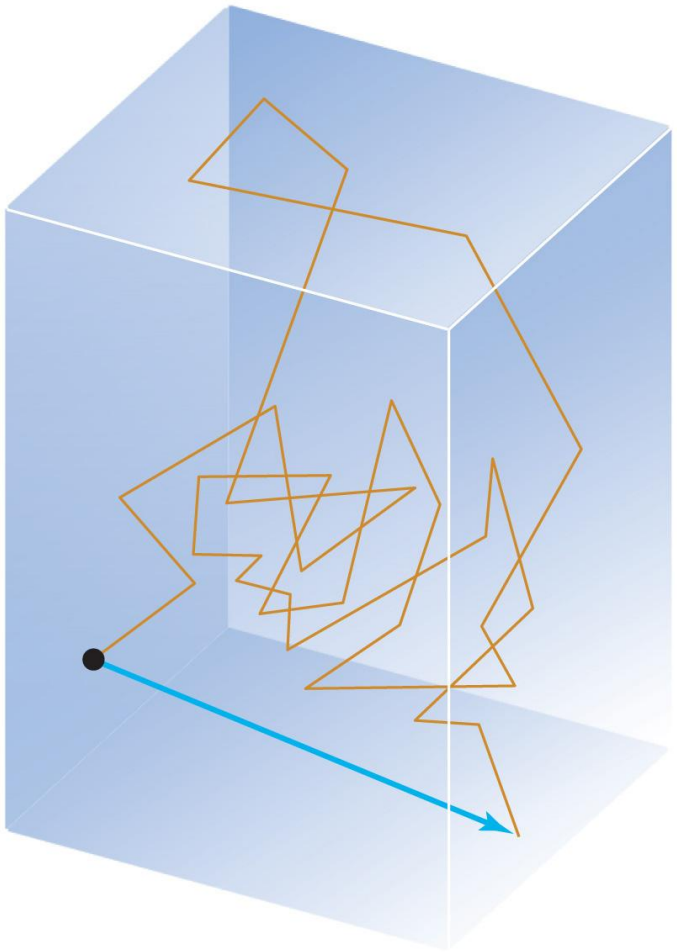


Two balloons are filled to the same volume, one with nitrogen and one with helium.

After 48 hours, the helium-filled balloon is smaller than the nitrogen-filled one because helium escapes faster than nitrogen.

Diffusion

Diffusion is the spread of one substance throughout a space or throughout a second substance.

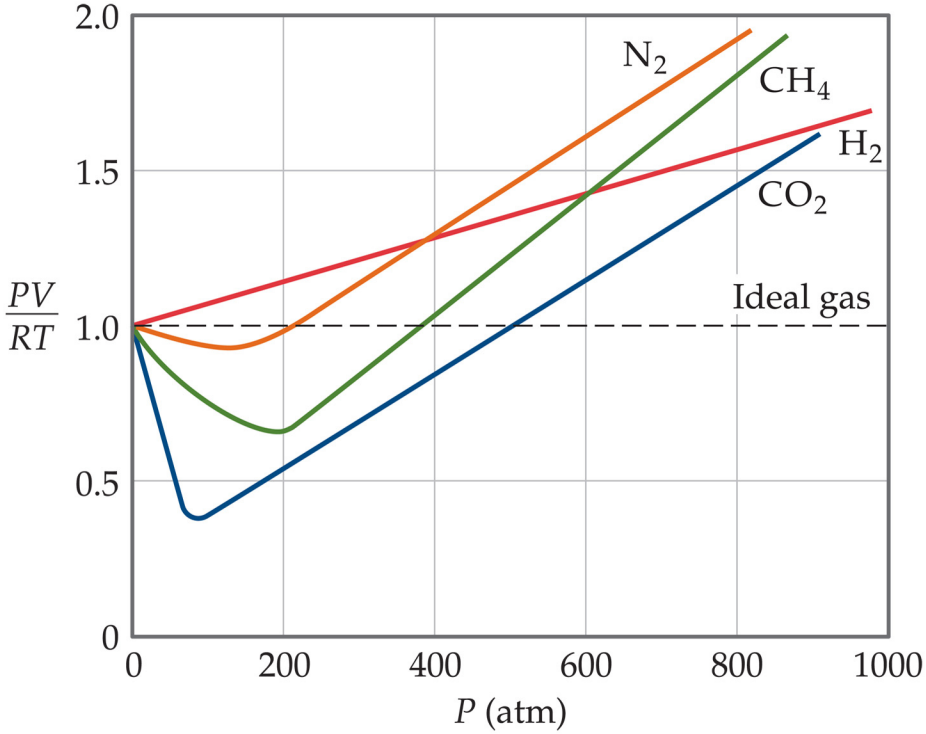


Diffusion

In general, we know that when a sample of gas is introduced to one part of a closed container, its molecules very quickly disperse throughout the container; this process by which molecules disperse in space in response to differences in concentration is called diffusion

The diffusion of gases allows the spread of the smell of a lot of things quickly. The smell of food, perfume, and fragrance of the incense stick diffuses in air.

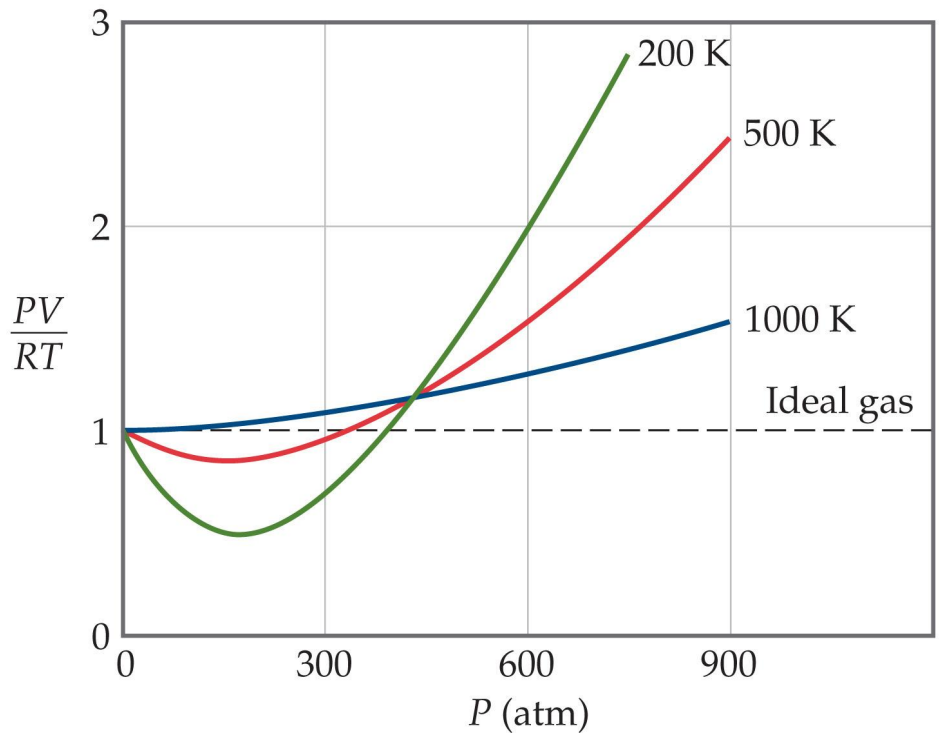
Real Gases



In the real world, the behavior of gases only conforms to the ideal-gas equation at relatively high temperature and low pressure.

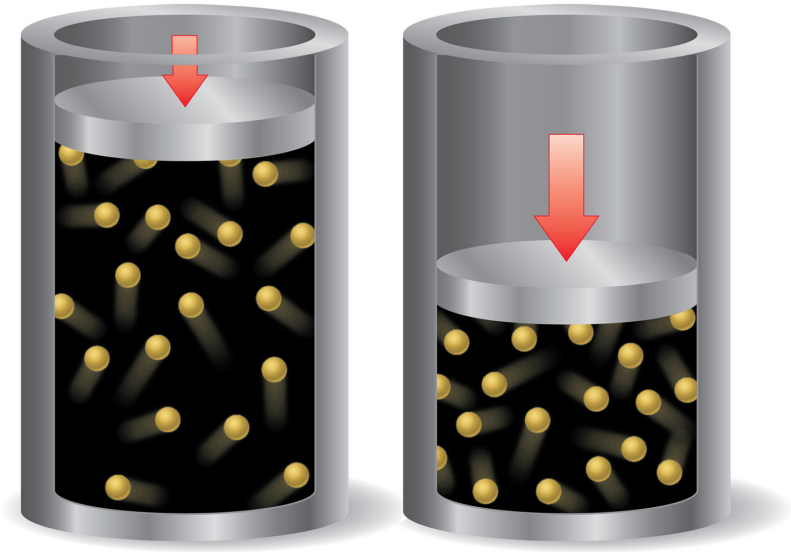
Real Gases

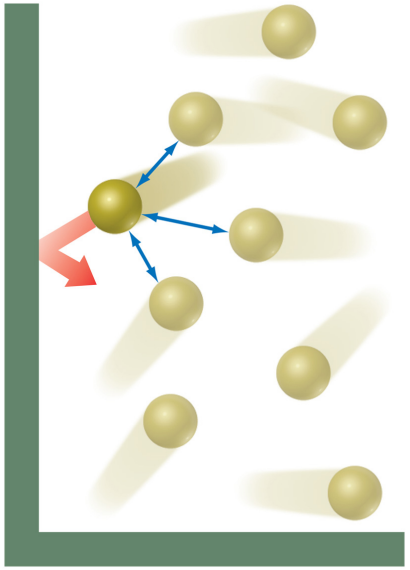
Even the same gas will show wildly different behavior under high pressure at different temperatures.



Gases

Deviations from Ideal Behavior





The assumptions made in the kinetic-molecular model (negligible volume of gas molecules themselves, no attractive forces between gas molecules, etc.) break down at high pressure and/or low temperature.

Corrections for Nonideal Behavior

- The ideal-gas equation can be adjusted to take these deviations from ideal behavior into account.
- The corrected ideal-gas equation is known as the van der Waals equation.

The van der Waals Equation

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

van der Waals Constants for Gas Molecules		
Substance	***a*** **(L^2-atm/mol^2)**	***b*** **(L/mol)**
He	0.0341	0.02370
Ne	0.211	0.0171
Ar	1.34	0.0322
Kr	2.32	0.0398
Xe	4.19	0.0510
H_2	0.244	0.0266
N_2	1.39	0.0391
O_2	1.36	0.0318
Cl_2	6.49	0.0562
H_2O	5.46	0.0305
CH_4	2.25	0.0428
CO_2	3.59	0.0427
CCl_4	20.4	0.1383

Characteristics Of Gas

- **Compressible**

The gas molecules have a lot of distance between each other that makes them compressible as they can move freely.

- **Expandable**

The gaseous state is also expandable because the free movement of gas molecules enables them to take over any shape on the basis of the container they are in, filling the volume of the container.

Low Density

They are a matter of lower density as compared to other states of matter, i.e. solids and liquids. The low density is the reason behind gas particles move randomly and quickly against each other without any steady position. Since there is too much distance between the molecules their collision does not meddle with their movement. The low density of gas also offers fluidity to this matter.

Diffusive

As said above, there is a huge distance between gas particles. Thus, two or more gases can combine easily and quickly, which is called the diffusion process. This combination creates a homogeneous mixture.

No Volume and Shape

The gaseous state neither has a fixed volume nor a fixed shape. It means that gas has no structure. The countless collisions of the molecules on the container walls result in the gaseous pressure. Thus, the shape and volume of a gas state both depend on the container holding it.